

MARQUETTE HARBOR, MI, USA

WMO: 997262

Lat: 46.546N

Lon: 87.379W

Elev: 615

StdP: 14.37

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.0	(c) -0.2	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 31.0	(k) 15.1	(l) 24.5	(m) 19.9	(n) 9.4	(o) 250	(p) 0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.9	(c) 81.6	(d) N/A	(e) 78.3	(f) N/A	(g) 75.3	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 9.5	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 24.5	(b) 20.1	(c) 17.0	(e) -7.9	(f) 88.4	(g) 5.0	(h) 3.2	(i) -11.5	(j) 90.8	(k) -14.5	(l) 92.7	(m) -17.3	(n) 94.5	(o) -21.0	(p) 96.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.2	18.6	18.8	28.0	38.3	48.1	57.3	65.7	65.3	58.5	47.0	34.7	24.2		
	DBStd	18.59	10.24	9.52	9.90	7.91	8.28	7.61	6.62	5.48	7.12	7.94	9.01	8.46		
	HDD50	4480	972	874	684	361	139	17	0	0	11	157	464	801		
	HDD65	8552	1437	1294	1147	801	527	253	74	64	218	561	910	1266		
	CDD50	1622	0	0	3	11	80	235	487	475	265	63	3	0		
	CDD65	218	0	0	0	0	3	21	96	74	21	3	0	0		
	CDH74	881	0	0	0	3	37	137	371	242	83	8	0	0		
	CDH80	151	0	0	0	0	7	28	66	35	14	1	0	0		
(14) Wind	WSAvg	6.9	8.3	7.8	7.9	7.9	7.3	5.5	5.4	5.5	6.2	6.7	7.5	7.5		
(15) Precipitation	PrecAvg	32.0	2.0	1.7	2.2	2.9	2.9	3.0	2.7	2.9	3.9	3.5	2.7	2.3		
	PrecMax	44.9	6.6	5.4	5.7	6.5	7.7	7.1	5.4	11.8	8.3	8.2	7.3	4.9		
	PrecMin	13.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.9	0.0	0.4		
	PrecStd	6.2	1.2	1.0	1.3	1.5	1.6	1.5	1.5	1.8	1.8	1.7	1.4	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.2	43.0	62.1	67.5	79.3	84.2	86.8	84.9	82.1	74.2	59.0	43.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	37.9	37.4	51.5	59.2	72.0	78.5	81.2	80.3	75.5	67.2	53.5	40.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	34.9	34.0	45.7	54.1	66.2	73.1	78.0	76.8	71.4	61.6	50.0	37.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	32.6	31.1	41.2	49.8	61.1	68.7	75.2	73.6	68.6	57.8	46.6	35.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	11.7	12.4	11.6	14.1	13.7	12.9	11.4	12.6	10.3	9.4	8.9		
		MCDBR	11.4	14.3	19.1	20.2	23.6	20.5	17.1	15.9	17.0	16.4	13.1	10.8		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.268	0.281	0.315	0.342	0.378	0.379	0.386	0.372	0.360	0.331	0.302	0.267		
	taud		2.430	2.381	2.371	2.391	2.366	2.390	2.397	2.437	2.469	2.523	2.520	2.484		
	Ebn at Noon		267	285	290	290	282	281	277	277	271	264	252	256		
	Edh at Noon		21	28	33	35	37	37	36	33	30	24	20	18		
(44) All-Sky Solar Radiation	RadAvg		329	593	1054	1411	1727	1872	1909	1619	1192	704	371	257		
	RadStd		25	80	100	148	122	142	100	103	80	84	48	22		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page